SecChaos-RL: Autonomous Threat-Aware Resilience Engineering Framework for Cloud Microservices

Closed-Loop Methodology: Continuous Chaos Injection → Telemetry Monitoring → Threat-Aware Neural Policy → Safe Multi-Objective Remediation

1. Security Chaos Engine (SCE)

Dynamic Fault Injection & Attacker Session Modeling

Standardized Security Fault Vectors

Poisson Process Arrivals: lambda_k in [0.05, 0.25]

F1: CVE-2023 Exploit Simulation (RCE, Injection)
F2: Privilege Escalation (Container Root Escape)
F3: Expired Service Certificates (TLS Handshake)
F4: Credential & Token Leaks (Vault Secret Leak)
F5: RBAC Role Manipulation (Scope Escalation)
F6: DoS Ingress Flooding (Thread Starvation)
F7: Security Config Drift (Permissive Policies)
F8: Malicious Package / Supply Chain Taint

Adversarial Attack Session Dynamics

- Concurrent attack sessions targeting vulnerable nodes
- Severity delta in [0.1, 1.0] grows with duration
- Triggers cascading timeouts across API graph

Stochastic Chaos Injection

2. Monitored Microservice Target System

DAG Dependency Topology G = (V, E) with Synchronous API Invocations

API Gateway — *Ingress / Route*
Auth Service — *OAuth / JWT Token*
Order Service — *Saga Coordinator*
Payment Svc — *Payment Gateway*
Inventory Svc — *Stock & Catalog*
Notify Service — *Async Kafka Queue*

Node Telemetry: Availability a_t(v) in [0,1], Risk r_t(v) in [0,1], Queue Depth

Prometheus & Istio Stream

3. Telemetry & State Monitor

Continuous Multi-Source Metric Aggregation & Observation Space

Telemetry Ingestion Pipeline

- Prometheus API: Latencies & P99 response times
- Istio Service Mesh: 5xx error rate spikes
- cAdvisor: Pod CPU & memory resource footprints
- Kubernetes API: Mitigation queue backlog
- SIEM / IDS: Global threat level L_threat in {0,1,2,3}
- tive mitigation status & rolling MTTR tracking

16-Dimensional State Vector s_t

Standardized Input Space for Neural Policy

State Vector s_t in R^16 (Normalized):
[0..5] Service security risks r(v_1..v_6)
[6] Mean ingress latency | [7] Service error rate
[8] CPU utilization | [9] Memory consumption
[10] Queue occupancy | [11] Threat severity L_threat
[12..14] Active actions | [15] Recovery step count

State Vector s_t

Metrics Feedback

Remediation Actuation

6. Multi-Objective SRE Reward

Balanced Trade-Off Formulation & PPO Buffer

Objective Functions

Scalar Reward Formula:
R_t = 5.0 * R_sec
+ 4.0 * A_t
+ 2.0 * R_mttr
- 0.5 * C_act
- 1.0 * C_wrong
Clipped to [-8.0, +8.0]

Trade-Off Drivers:
- Signed Risk Red R_sec
- Availability SLA A_t in [0,1]
- Recovery Speed R_mttr
- Action Churn Cost C_act
- Wrong Action Penalty C_wrong

PPO Loss Function:
L = L_clip - c1*L_VF
+ c2*S[pi] - c3*L_aux

5. Safety Gate & Cloud Actuator

Operational Validation (Alg. 1) & Kubernetes Operators

Safety Validation Gate

1. Algorithmic Guardrails (Alg. 1):
   Precondition check against DAG G
2. Queue Redundancy Suppression:
   Drops duplicate identical actions
3. Transient Failure Model: eps_queue = 8%
4. False Mitigation Rate (FMR) Tracking

Cloud-Native Actuators A in {0..8}

Decision Latency: 0.44 +/- 0.15 ms

a=0: No-Op / Standby (Resource Conserve)
a=1: cert-manager Secret Renewal
a=2: Pod Rolling Patch Hotfix
a=3: HashiCorp Vault Secret Rotation
a=4: RBAC RoleBinding Lockdown
a=5: Redis Session Cache Purge & MFA
a=6: Container Image Tag Patch Update
a=7: ValidatingAdmission Policy Strip
a=8: Helm / ReplicaSet Rollback

Action Proposal a_t

PPO Advantage & Loss Gradient Backprop

4. SecChaos-RL Neural Policy Network Architecture

Threat-Aware Deep Multi-Objective Actor-Critic with Multi-Head Self-Attention

1. Linear Feature Projection Layers

e_t^(i) = W_e^(i) s_t^(i) + b_e^(i) in R^32 for each feature dimension i in {0, ..., 15}

2. Threat-Aware Feature Gating

*g_t^(i) = e_t^(i) * (1.0 + L_threat) for security channels (i <= 4); unamplified for QoS channels*

Dynamically amplifies critical security features under active attack conditions

3. Multi-Head Self-Attention (MHSA) with Residual Connection

h=4 Attention Heads, d_k=32, Scaled Dot-Product: Softmax(Q K^T / sqrt(32)) V

Captures non-linear cross-service dependencies and cascading failure paths
LayerNorm(x + MHSA(x)) stabilizes shared latent representation z_t in R^128

Policy / Actor Head

pi_theta(a | s)

MLP 128->64->9
Threat Temp tau
Clipped PPO loss

Critic Baseline Head

V_phi(s)

MLP 128->64->1
Value function
GAE Advantage A_t

Auxiliary Risk Head

r_hat_(t+1) in R^6

MLP 128->64->6
MSE Risk Loss
Future hazard est.